Molecular genetic characterization of 10 unrelated families with autosomal-recessive inherited retinal dystrophies: a combined exome and targeted-gene analysis, including two Ashkenazi Jewish pedigrees with *DHDDS*-related retinitis pigmentosa

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ABSTRACT

Purpose. To identify the molecular cause of disease in 10 unrelated probands with clinically diagnosed autosomal-recessive inherited retinal dystrophy (AR-IRD), and to characterize two Ashkenazi Jewish families with retinitis pigmentosa type 59 (RP59) caused by the founder *DHDDS* variant c.124A>G (p.Lys42Glu). **Methods.** Whole-exome sequencing (WES) was performed in eight families (F1–F8). In two additional Ashkenazi Jewish pedigrees (F9, F10) with a strong prior probability of *DHDDS*-related disease, targeted Sanger sequencing of all 12 coding exons of *DHDDS* was performed as a first-tier test. Candidate variants were filtered against the gnomAD (v2.0.2) database, scored with CADD v1.3, and classified per ACMG/AMP 2015 guidelines. Segregation was tested by Sanger sequencing in all available affected and unaffected first-degree relatives. **Results.** A definitive molecular diagnosis was established in all 10 families. Causal genes were *USH2A* (F1), *EYS* (F2), *RPE65* (F3), *CRB1* (F4), *CNGB1* (F5), *PDE6A* (F6), *ABCA4* (F7), *NPHP1* (F8), and *DHDDS* (F9, F10). Two families (F7 and F8) required formal segregation analysis to resolve between competing candidate genes identified by WES. The two Ashkenazi Jewish probands (F9, F10) were both homozygous for *DHDDS* c.124A>G, contributing a combined total of **four pathogenic alleles** across the two families. **Conclusions.** A first-tier targeted test for the Ashkenazi founder allele *DHDDS* c.124A>G remains a cost-effective approach in this population and reliably identifies biallelic carriers. WES is required to disambiguate multi-gene candidate findings in non-founder families, as illustrated by F7 and F8 in this cohort.

Keywords: inherited retinal dystrophy; retinitis pigmentosa; Leber congenital amaurosis; whole-exome sequencing; *DHDDS*; Ashkenazi Jewish founder mutation; ACMG variant classification.

1. INTRODUCTION

Inherited retinal dystrophies (IRDs) are a clinically and genetically heterogeneous group of disorders that together affect approximately 1 in 3,000 individuals worldwide (Hartong et al., 2006). More than 270 genes have been associated with non-syndromic and syndromic IRDs to date (RetNet, accessed January 2017), and autosomal-recessive (AR) inheritance accounts for an estimated 50–60% of cases. Despite advances in next-generation sequencing, between 30% and 40% of clinically diagnosed AR-IRD probands remain genetically unsolved after a single round of targeted-panel testing, motivating the use of whole-exome sequencing (WES). Targeted founder-allele testing remains, however, the most cost-effective first-tier strategy in populations with well-characterized recurrent variants. The Ashkenazi Jewish (AJ) population is one such example: the *DHDDS* c.124A>G (p.Lys42Glu) variant has an estimated carrier frequency of 1 in 322 in the AJ population (Züchner et al., 2011) and accounts for essentially all reported AJ cases of RP59. In the present study we report molecular findings in 10 unrelated AR-IRD probands ascertained at the Northgate Eye Institute between June 2014 and November 2016. Eight families underwent WES; two AJ families with strong prior probability of *DHDDS*-related disease underwent first-tier targeted Sanger sequencing of *DHDDS*. Two of the eight WES families (F7 and F8) returned multiple candidate findings, illustrating that careful segregation analysis is required to distinguish the causal variant from incidental biallelic findings in unrelated genes.

2. MATERIALS AND METHODS

2.1 Subjects and clinical evaluation

Ten unrelated probands (F1–F10) and 27 first-degree relatives were enrolled under IRB protocol NGEI-2014-0431. All affected individuals underwent best-corrected visual acuity testing, dilated fundus examination, full-field electroretinography (ffERG) per ISCEV 2008 standards, spectral-domain optical coherence tomography (SD-OCT, Heidelberg Spectralis), and Goldmann visual field testing. Age at onset (AAO) was defined as the age at which the proband or parent first reported night blindness or visual disturbance. The clinical cohort comprised 13 affected individuals across the 10 pedigrees (Table 1).

2.2 Whole-exome sequencing (F1–F8)

Genomic DNA was extracted from peripheral blood using the QIAamp DNA Blood Maxi Kit (Qiagen). Exome capture used the SureSelect Human All Exon V6 kit (Agilent, 60 Mb target). Libraries were sequenced on an Illumina HiSeq 4000 in 2 × 100 bp paired-end mode. Mean on-target coverage across the eight WES samples was 112.3× (range 98.4×–127.8×), with 96.1% of target bases covered at ≥20× (Table 3). Reads were aligned to GRCh37 using BWA-MEM v0.7.15, duplicates marked with Picard MarkDuplicates v2.6.0, and variants called with GATK HaplotypeCaller v3.7. Variants were annotated with ANNOVAR (2016Feb01 build) and Variant Effect Predictor v87.

2.3 Targeted *DHDDS* sequencing (F9, F10)

For the two Ashkenazi Jewish pedigrees, all 12 coding exons of *DHDDS* (NM_024887.3) and ≥50 bp of flanking intronic sequence were amplified by PCR (primers in Supplementary Table S2) and Sanger-sequenced on an ABI 3730xl. Sequences were aligned to the *DHDDS* reference (NM_024887.3) using Mutation Surveyor v5.0.1. The known founder variant c.124A>G (p.Lys42Glu) was confirmed by independent forward and reverse reads in every carrier and homozygote.

2.4 Variant filtering and classification

Candidate variants were retained if they (i) had a minor allele frequency < 0.005 in gnomAD v2.0.2 (all populations), (ii) were predicted to be protein-altering (missense, nonsense, frameshift, canonical splice, or in-frame indel), (iii) had a CADD v1.3 phred score ≥ 20, and (iv) were present in a gene previously associated with an IRD phenotype consistent with the proband (RetNet, January 2017 release). Variants were classified per ACMG/AMP 2015 guidelines as benign (B), likely benign (LB), variant of uncertain significance (VUS), likely pathogenic (LP), or pathogenic (P). Only P and LP variants are reported as causal in Section 3.

3. RESULTS

3.1 Cohort summary

Table 1 summarizes the 10 families. Thirteen affected individuals were studied (5 female, 8 male); mean age at study enrollment was 28.4 years (range 4–55). Mean age at reported onset of visual symptoms (AAO) was 14.9 years (range 2–28). Six families presented with non-syndromic retinitis pigmentosa (RP); two with Leber congenital amaurosis (LCA); one with Stargardt disease (STGD, F7); one with Senior–Løken syndrome (RP plus nephronophthisis, F8). The two Ashkenazi Jewish families (F9, F10) had non-syndromic RP59.

Family	Ancestry	Method	Affected (n)	Sex (proband)	AAO (yr)	Phenotype
F1	European	WES	1	M	14	RP
F2	European	WES	2	F	18	RP
F3	South Asian	WES	1	M	2	LCA
F4	European	WES	2	F	3	LCA
F5	Middle Eastern	WES	1	M	22	RP
F6	African	WES	1	M	16	RP
F7	European	WES	2	F	12	STGD
F8	South Asian	WES	1	M	9	RP + NPH (Senior–Løken)
F9	Ashkenazi Jewish	Targeted DHDDS	1	F	25	RP59
F10	Ashkenazi Jewish	Targeted DHDDS	1	M	28	RP59

Table 1. Demographic and clinical summary of the 10 unrelated AR-IRD families. AAO, age at onset; RP, retinitis pigmentosa; LCA, Leber congenital amaurosis; STGD, Stargardt disease; NPH, nephronophthisis. F9 and F10 are Ashkenazi Jewish probands tested by first-tier targeted Sanger sequencing of *DHDDS*.

3.2 WES sequencing performance (F1–F8)

Mean on-target coverage, percent of target bases covered ≥ 10× and ≥ 20×, total mapped reads, and percent duplicate reads are summarized for each WES sample in Table 3. All eight WES samples passed quality thresholds (≥ 95% of target bases ≥ 20×).

Family	Mean cov. (×)	% ≥ 10×	% ≥ 20×	Mapped reads (M)	% duplicates	Ti/Tv
F1	118.7	98.4	96.3	84.2	11.8	2.04
F2	102.5	97.9	95.1	76.4	13.2	2.07
F3	98.4	97.6	95.0	71.9	12.7	2.05
F4	127.8	98.7	97.2	92.6	10.9	2.06
F5	114.1	98.2	96.0	82.0	11.4	2.03
F6	109.6	98.0	95.7	79.8	12.1	2.05
F7	121.3	98.5	96.8	86.5	11.6	2.06
F8	106.0	97.8	95.4	78.1	12.0	2.04
Mean	**112.3**	**98.1**	**96.1**	**81.4**	**11.96**	**2.05**

Table 3. WES sequencing performance metrics for the eight WES-tested families. Coverage is reported across the SureSelect V6 target (60 Mb). Ti/Tv, transition-to-transversion ratio of high-quality SNVs (within expected range 2.0–2.1).

3.3 Causal variants across the cohort

A definitive molecular diagnosis was established in every family. Table 2 lists each pathogenic or likely-pathogenic variant identified, with zygosity, gnomAD v2.0.2 allele frequency, CADD v1.3 phred score, and final ACMG/AMP classification. Across all 10 families, a total of 16 distinct pathogenic/likely-pathogenic alleles were observed (counting compound heterozygotes as two alleles per proband and homozygotes as two alleles per proband). The two Ashkenazi Jewish probands (F9 and F10) each contributed two alleles of the founder *DHDDS* c.124A>G variant (homozygous), for a subtotal of **four alleles in the Ashkenazi *DHDDS* subcohort** (see Section 3.5).

Fam.	Gene	Transcript	cDNA	Protein	Zyg.	gnomAD AF	CADD	ACMG
F1	USH2A	NM_206933.2	c.2299delG	p.Glu767Serfs*21	hom	$1.10\times10^{\blacksquare 3}$	33	P
F2	EYS	NM_001142800.1	c.4350_4356del	p.Ile1451Profs*3	het	$3.20\times10^{\blacksquare\blacksquare}$	32	P
F2	EYS	NM_001142800.1	c.6557G>A	p.Gly2186Glu	het	$8.10\times10^{\blacksquare\blacksquare}$	28	LP
F3	RPE65	NM_000329.2	c.271C>T	p.Arg91Trp	hom	$4.40\times10^{\blacksquare\blacksquare}$	29	P
F4	CRB1	NM_201253.2	c.2843G>A	p.Cys948Tyr	hom	$2.10\times10^{\blacksquare\blacksquare}$	30	P
F5	CNGB1	NM_001297.4	c.2957A>T	p.Asn986Ile	hom	$9.00\times10^{\blacksquare\blacksquare}$	27	LP
F6	PDE6A	NM_000440.2	c.1957C>T	p.Arg653*	het	$2.30\times10^{\blacksquare\blacksquare}$	37	P
F6	PDE6A	NM_000440.2	c.2053G>A	p.Val685Met	het	$1.40\times10^{\blacksquare\blacksquare}$	26	LP
F7	ABCA4	NM_000350.2	c.5882G>A	p.Gly1961Glu	het	$4.10\times10^{\blacksquare 3}$	24	P
F7	ABCA4	NM_000350.2	c.6079C>T	p.Leu2027Phe	het	$5.00\times10^{\blacksquare\blacksquare}$	26	LP
F8	NPHP1	NM_000272.3	exon 2–6 del	p.(?)	hom	$1.50\times10^{\blacksquare\blacksquare}$	n/a	P
F9	DHDDS	NM_024887.3	c.124A>G	p.Lys42Glu	hom	$1.55\times10^{\blacksquare 3}$ (AJ 0.31%)	25	P
F10	DHDDS	NM_024887.3	c.124A>G	p.Lys42Glu	hom	$1.55\times10^{\blacksquare 3}$ (AJ 0.31%)	25	P

Table 2. Causal pathogenic (P) and likely-pathogenic (LP) variants in the 10 families. Zygosity: hom, homozygous; het, heterozygous. gnomAD allele frequencies are from v2.0.2 (all populations); for *DHDDS* c.124A>G the Ashkenazi-Jewish-specific frequency is also given in parentheses. CADD scores are v1.3 phred-scaled. The two highlighted rows (F9, F10) form the Ashkenazi *DHDDS* subcohort and each contribute two pathogenic alleles to the combined total of four (see Table 5).

3.4 Per-family findings

Family F1 (European; *USH2A* c.2299delG homozygous; RP). The proband (II-1) is a 32-year-old man with non-syndromic RP and reported AAO of 14 years. ffERG showed non-detectable rod responses and severely reduced cone responses (b-wave 18 μV; lower limit of normal 110 μV). SD-OCT showed loss of the ellipsoid zone beyond 8.4° eccentricity bilaterally. WES identified the well-characterized *USH2A* c.2299delG (p.Glu767Serfs*21) variant in homozygous state (read depth 84×, allele balance 0.98). Both parents were confirmed as heterozygous carriers (Sanger). Hearing was normal on age-appropriate audiometric testing; the proband was therefore classified as having non-syndromic *USH2A*-related RP rather than Usher syndrome. **Allele count contributed: 2 (homozygous).**

Family F2 (European; *EYS* compound heterozygous; RP). Two affected sisters (II-1, age 24; II-2, age 21) and one unaffected brother (II-3, age 19). AAO 18 years in II-1; 19 years in II-2. WES of II-1 identified compound heterozygous variants in *EYS*: c.4350_4356del (p.Ile1451Profs*3; paternal) and c.6557G>A (p.Gly2186Glu; maternal). Segregation in II-2 confirmed the same compound state (read depths 64× and 71× respectively). The unaffected brother II-3 carried only the maternal allele. **Allele count contributed: 2.**

Family F3 (South Asian, consanguineous; *RPE65* c.271C>T homozygous; LCA). Proband II-1 is a 4-year-old boy with severe early-onset visual impairment (AAO < 6 months by parental report, with nystagmus first noted at 8 weeks). Fundus examination showed pigmentary changes and a blunted foveal reflex. ffERG was non-detectable for both rod and cone responses. WES revealed a homozygous c.271C>T (p.Arg91Trp) variant in *RPE65* (read depth 96×). The parents are first cousins and both confirmed heterozygous. **Allele count contributed: 2.**

Family F4 (European; *CRB1* c.2843G>A homozygous; LCA). Two affected sisters (II-1, age 7; II-2, age 4). Both presented with nystagmus before 12 months of age and oculo-digital signs. WES identified a homozygous *CRB1* c.2843G>A (p.Cys948Tyr) variant, a recurrent pathogenic allele previously associated with LCA8 (Lotery et al., 2001). Read depths were 78× (II-1) and 81× (II-2). Parents were both heterozygous and clinically unaffected. **Allele count contributed: 2.**

Family F5 (Middle Eastern, consanguineous; *CNGB1* c.2957A>T homozygous; RP). Proband II-1, a 31-year-old man, AAO 22 years. ffERG rod responses non-detectable; cone b-wave 42 μV. WES identified a homozygous missense variant in *CNGB1* (c.2957A>T, p.Asn986Ile; read depth 67×, CADD 27). Both consanguineous parents (first cousins) heterozygous. **Allele count contributed: 2.**

Family F6 (African; *PDE6A* compound heterozygous; RP). Proband II-2, age 26, AAO 16 years. WES identified *PDE6A* c.1957C>T (p.Arg653*; nonsense, CADD 37, paternal) and c.2053G>A (p.Val685Met; missense, CADD 26, maternal). Both parents confirmed as single-allele carriers; unaffected sister II-1 carried only the maternal allele. **Allele count contributed: 2.**

Family F7 (European; resolved as *ABCA4*; STGD)

Family F7 illustrates the importance of formal segregation analysis when WES returns more than one plausible biallelic finding. The proband II-1 (age 19, AAO 12 years) and her affected sister II-2 (age 16, AAO 13 years) had clinical features consistent with juvenile-onset Stargardt disease: bilateral central scotomata, foveal atrophy on SD-OCT, and characteristic flecks on fundus autofluorescence. Their unaffected brother II-3 (age 22) was included as a within-family control. Initial WES analysis of II-1 returned **two candidate biallelic findings** that each individually passed our filtering criteria (Supplementary Table S1):

Gene	Variant	Zygosity in II-1	gnomAD AF	CADD	Initial ACMG
ABCA4	c.5882G>A (p.Gly1961Glu) [maternal] + c.6079C>T (p.Leu2027Phe) [paternal]	compound het	4.1×10■³ / 5.0×10■■	24 / 26	P / LP
USH2A	c.11864G>A (p.Trp3955*) heterozygous + c.2276G>T (p.Cys759Phe) heterozygous	compound het (apparent)	2.6×10■■ / 4.3×10■³	39 / 24	P / LP

Supplementary Table S1. Two competing biallelic candidate findings in F7 proband II-1, prior to segregation testing.

Segregation testing distinguished the two candidates. The affected sister II-2 carried **both** *ABCA4* alleles in compound state (c.5882G>A maternal + c.6079C>T paternal, confirmed by independent Sanger reactions) but carried **only the maternal *USH2A* c.2276G>T allele** and not the paternal *USH2A* c.11864G>A allele (Sanger trace negative; read depth on parental sample 41×). The unaffected brother II-3 carried only the maternal *ABCA4* c.5882G>A allele (a single heterozygous carrier state). Phenotypically, neither sister had any feature of Usher syndrome on age-appropriate audiometric testing or vestibular review.

We therefore concluded that **the causal gene in family F7 is *ABCA4***, with the compound heterozygous combination c.5882G>A (p.Gly1961Glu) / c.6079C>T (p.Leu2027Phe). The apparent biallelic *USH2A* finding in the proband did not segregate and was incidental; the *USH2A* c.11864G>A allele is reclassified as a coincidental carrier finding and the *USH2A* c.2276G>T variant as a VUS-carrier finding. **Allele count contributed by F7 to the causal-variant tally: 2 (the two *ABCA4* alleles).** *USH2A* alleles are not counted toward the causal tally.

Family F8 (South Asian, consanguineous; resolved as *NPHP1*; Senior–Løken)

F8 is a consanguineous South Asian family in which the proband II-2 (age 11, AAO 9 years for visual symptoms) presented with retinitis pigmentosa together with progressive renal impairment (eGFR 48 mL/min/1.73 m^2 at age 10; renal biopsy consistent with nephronophthisis). His unaffected older sister II-1 (age 14) was clinically and renally unremarkable (eGFR 112 mL/min/1.73 m^2). WES of the proband returned **two candidate biallelic findings** in ciliopathy-associated genes (Supplementary Table S2):

Gene	Variant	Zygosity in II-2	gnomAD AF	CADD	Initial ACMG
IQCB1	c.1090C>T (p.Arg364Cys) homozygous	hom	1.8×10■^{2}	23	VUS
NPHP1	exon 2–6 multi-exon deletion (~ 87 kb)	hom (LR-PCR confirmed)	1.5×10■■	n/a (CNV)	P

Supplementary Table S2. Two competing biallelic candidate findings in F8 proband II-2, prior to segregation and CNV confirmation.

The *IQCB1* c.1090C>T variant initially appeared compelling because (i) biallelic *IQCB1* variants cause Senior–Løken syndrome type 5, matching the proband's phenotype, and (ii) the variant was homozygous in this consanguineous pedigree. However, two pieces of evidence reclassified *IQCB1* c.1090C>T as **benign**: (a) The variant is present in gnomAD v2.0.2 at an overall allele frequency of 1.8×10■^{2} (South Asian sub-population: 2.4×10■^{2}), with 7 homozygous individuals in gnomAD, which exceeds the BS1/BA1 thresholds for a Senior–Løken-causing variant. (b) The unaffected older sister II-1 was also homozygous for *IQCB1* c.1090C>T (confirmed by Sanger), excluding it as the cause of disease in this family.

The *NPHP1* exon 2–6 deletion was suggested by WES read-depth analysis (mean coverage across *NPHP1* exons 2–6 was 0× in II-2 vs 78.1× mean genome-wide) and confirmed by long-range PCR using primers spanning the predicted breakpoints (forward 5'-CCTGAAGTCAGTATGCCTGTAGCTT-3'; reverse 5'-AGTGATCACAGCATTGCTCAAGGAA-3'; expected wild-type product 6.2 kb; mutant breakpoint product 2.4 kb). Both parents carried the heterozygous deletion; the unaffected sister II-1 was heterozygous (a carrier) but did not carry it in homozygous state.

We therefore concluded that **the causal gene in family F8 is *NPHP1***, with a homozygous exon 2–6 deletion. The *IQCB1* c.1090C>T finding was reclassified as a benign incidental finding. **Allele count contributed by F8 to the causal-variant tally: 2 (the two deleted *NPHP1* alleles).** *IQCB1* alleles are not counted toward the causal tally.

3.5 Ashkenazi Jewish *DHDDS* subcohort (F9, F10)

Two unrelated Ashkenazi Jewish probands (F9 and F10) were ascertained from the Northgate Eye Institute clinical service with a phenotype compatible with RP59 and a self-reported four-grandparent Ashkenazi background. Because the founder *DHDDS* c.124A>G (p.Lys42Glu) allele accounts for essentially all reported AJ RP59 (Züchner et al., 2011), targeted Sanger sequencing of all 12 coding exons of *DHDDS* (NM_024887.3) was performed as a first-tier test rather than WES.

Family F9. Proband II-1 is a 37-year-old woman with AAO 25 years. Visual acuity at presentation 20/40 OD, 20/50 OS; ffERG rod responses non-detectable, cone b-wave amplitude 31 μV. Targeted Sanger sequencing of *DHDDS* identified the c.124A>G (p.Lys42Glu) variant in **homozygous** state, confirmed on both forward and reverse traces. Her unaffected mother and father were each confirmed as heterozygous carriers (single-allele, by Sanger). **Allele count contributed: 2.**

Family F10. Proband II-1 is a 44-year-old man with AAO 28 years. Visual acuity 20/60 OD, 20/80 OS; central visual fields preserved to 12°; ffERG non-detectable rod responses; cone responses severely reduced (b-wave 22 μV). Targeted Sanger sequencing of *DHDDS* identified c.124A>G (p.Lys42Glu) in **homozygous** state. The proband's unaffected adult son and daughter were each confirmed as heterozygous carriers, consistent with obligate transmission. **Allele count contributed: 2.**

Combined Ashkenazi *DHDDS* tally. Together, the two Ashkenazi Jewish families F9 and F10 contributed **four pathogenic alleles of *DHDDS* c.124A>G** (two homozygous probands × two alleles each) to the causal-variant tally of this cohort. No carrier (heterozygous) parent or sibling alleles are counted toward this tally. The combined four-allele subtotal is summarized in Table 5.

Family	Proband	Gene	Variant	Zygosity	Alleles contributed
F9	II-1	DHDDS	c.124A>G (p.Lys42Glu)	homozygous	2
F10	II-1	DHDDS	c.124A>G (p.Lys42Glu)	homozygous	2
				Total (F9 + F10):	**4**

Table 5. Pathogenic-allele tally in the Ashkenazi Jewish *DHDDS* subcohort. Each homozygous proband contributes two alleles of the founder c.124A>G (p.Lys42Glu) variant. **Combined total across F9 and F10 = 4 pathogenic alleles.**

3.6 Cohort-wide allele tally

Across all 10 families, a total of 20 pathogenic/likely-pathogenic *causal* alleles were identified, distributed as 2 alleles per proband (every proband was either homozygous or compound heterozygous for two causal alleles in the same gene). The four alleles from the Ashkenazi *DHDDS* subcohort (F9 + F10) account for 20% of the cohort-wide causal-allele tally.

4. DISCUSSION

We report molecular diagnoses in 10 unrelated AR-IRD probands using a combined strategy of whole-exome sequencing (n = 8) and population-targeted Sanger sequencing of a single founder gene (n = 2). All ten families were solved. Two of the eight WES-tested families (F7 and F8) returned more than one candidate biallelic finding in genes individually plausible on the basis of phenotype, and required segregation analysis (and, in the case of F8, long-range PCR for breakpoint confirmation) to resolve the causal allele. The two targeted-DHDDS families (F9, F10), in contrast, each returned a single unambiguous homozygous result for the Ashkenazi founder allele c.124A>G.

The F7 multi-gene ambiguity is instructive. The *ABCA4* c.5882G>A (p.Gly1961Glu) allele is well-known as the most common pathogenic *ABCA4* allele in European-ancestry Stargardt cohorts (gnomAD AF $4.1\times10^{\blacksquare 3}$), and would be flagged by virtually any exome pipeline. In our F7 proband, however, WES also returned an apparent compound heterozygous *USH2A* state. Without segregation testing in the affected sister II-2 and the unaffected brother II-3, both findings would have appeared individually causal. Only segregation (and the absence of any Usher syndrome features) supported *ABCA4* as the causal gene. The *USH2A* alleles in F7 are incidental carrier findings and are not included in the Table 2 causal-variant tally.

The F8 multi-gene ambiguity is similarly informative. An apparent homozygous *IQCB1* missense variant initially looked compelling because biallelic *IQCB1* loss-of-function causes Senior–Løken syndrome (matching the proband's combined retinal and renal phenotype) and the proband's parents were first cousins. However, the homozygous presence of the same variant in the proband's clinically unaffected sister, together with the high overall gnomAD frequency ($1.8\times10^{\blacksquare 2}$; 7 homozygotes), reclassified *IQCB1* c.1090C>T as benign per BS1/BS2/BS4 ACMG/AMP criteria. The *NPHP1* exon 2–6 homozygous deletion is the causal allele in F8.

The Ashkenazi *DHDDS* subcohort (F9, F10) supports the continued value of population-specific first-tier targeted testing. In a population with a known high-frequency founder allele, sequencing all 12 coding exons of *DHDDS* by Sanger costs approximately one-fifteenth of WES per sample and returned an unambiguous biallelic result in both probands. Combined across the two families, four pathogenic *DHDDS* c.124A>G alleles were identified, all in homozygous state in the probands, with the expected obligate-carrier pattern in parents and adult offspring.

5. CONCLUSIONS

A first-tier targeted Sanger test for the Ashkenazi founder allele *DHDDS* c.124A>G remains the most cost-effective approach to *DHDDS*-related RP59 in self-identified Ashkenazi Jewish probands; in our two such families it returned a definitive diagnosis in both probands, contributing a combined total of four pathogenic alleles across the subcohort. For non-founder AR-IRD probands, whole-exome sequencing combined with careful segregation analysis remains essential to resolve multi-gene candidate findings, as illustrated by F7 (*ABCA4* vs *USH2A*) and F8 (*NPHP1* vs *IQCB1*) in this cohort.

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